



Software Firewall Implementation & Wazuh Integration

Task 2 Booklet

SOC Winter Internship Program 2026

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January 2026

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Objectives

The goal of this task is to implement a software-based firewall (pfSense) and integrate it with Wazuh SIEM for centralized security monitoring. This task focuses on:

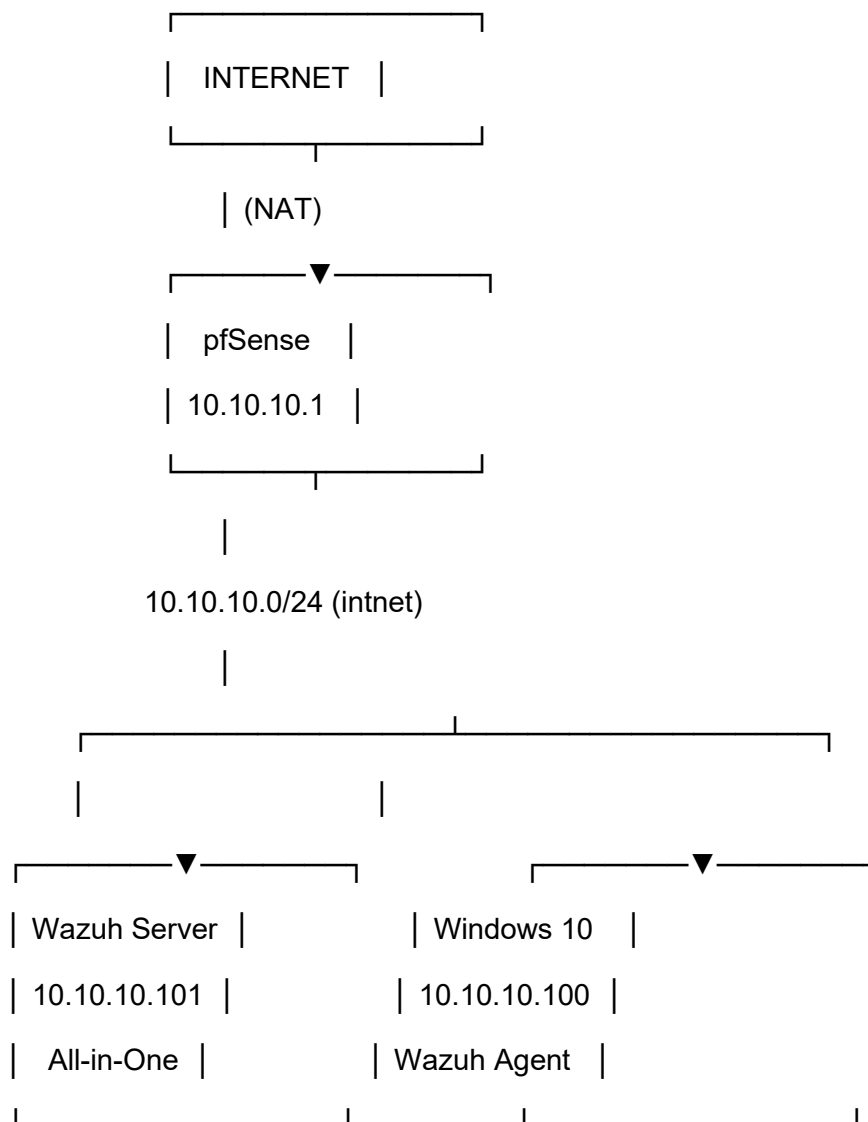
- Blocking traffic from specific countries (China, Russia) using GeoIP
- Restricting access to specified websites using DNS blacklisting
- Creating administrator privilege rules to control access
- Monitoring firewall logs and file integrity through Wazuh
- Generating security reports based on observed traffic

Additionally, File Integrity Monitoring (FIM) was configured on the Windows client to detect unauthorized file changes in real-time, providing early indicators of compromise or malicious activity.

Tools & Environment Setup

Step 1: Lab Architecture

The following virtual environment was created using Oracle VM VirtualBox:



Step 2: Virtual Machine Resource Allocation

Component	CPU	RAM	Storage	Network
pfSense Firewall	1 Core	2 GB	20 GB	NAT + Internal
Wazuh Server	2 Cores	4 GB	50 GB	Internal
Windows 10 Client	2 Cores	4 GB	50 GB	Internal

Step 3: Network Configuration

Device	IP Address	Role
pfSense LAN	10.10.10.1	Gateway/Firewall
Wazuh Server	10.10.10.101	SIEM Server
Windows 10	10.10.10.100	Client + Wazuh Agent

Network Settings:

- pfSense Adapter 1: NAT (WAN - Internet access)
- pfSense Adapter 2: Internal Network "intnet" (LAN)
- Wazuh Adapter 1: Internal Network "intnet"
- Windows 10 Adapter 1: Internal Network "intnet"

pfSense Firewall Installation

Step 4: Installing pfSense

The pfSense Community Edition ISO was downloaded and installed on a virtual machine.

Download Link: <https://www.pfsense.org/download/>

Installation Steps:

1. Created new VM in VirtualBox
2. Attached pfSense ISO to virtual optical drive
3. Booted VM and followed installation wizard
4. Configured WAN interface (em0) with NAT
5. Configured LAN interface (em1) with IP 10.10.10.1/24
6. Enabled DHCP server on LAN (range: 10.10.10.100 - 10.10.10.200)

```
pfsense-firewall [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Starting /usr/local/etc/rc.d/pfb_filter.sh...done.
pfSense 2.8.1-RELEASE amd64 20251215-1731
bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 5696f831cd2f94e7b595

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe7d:a054/64
LAN (lan) -> em1 -> v4: 10.10.10.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Step 5: Accessing pfSense Web Dashboard

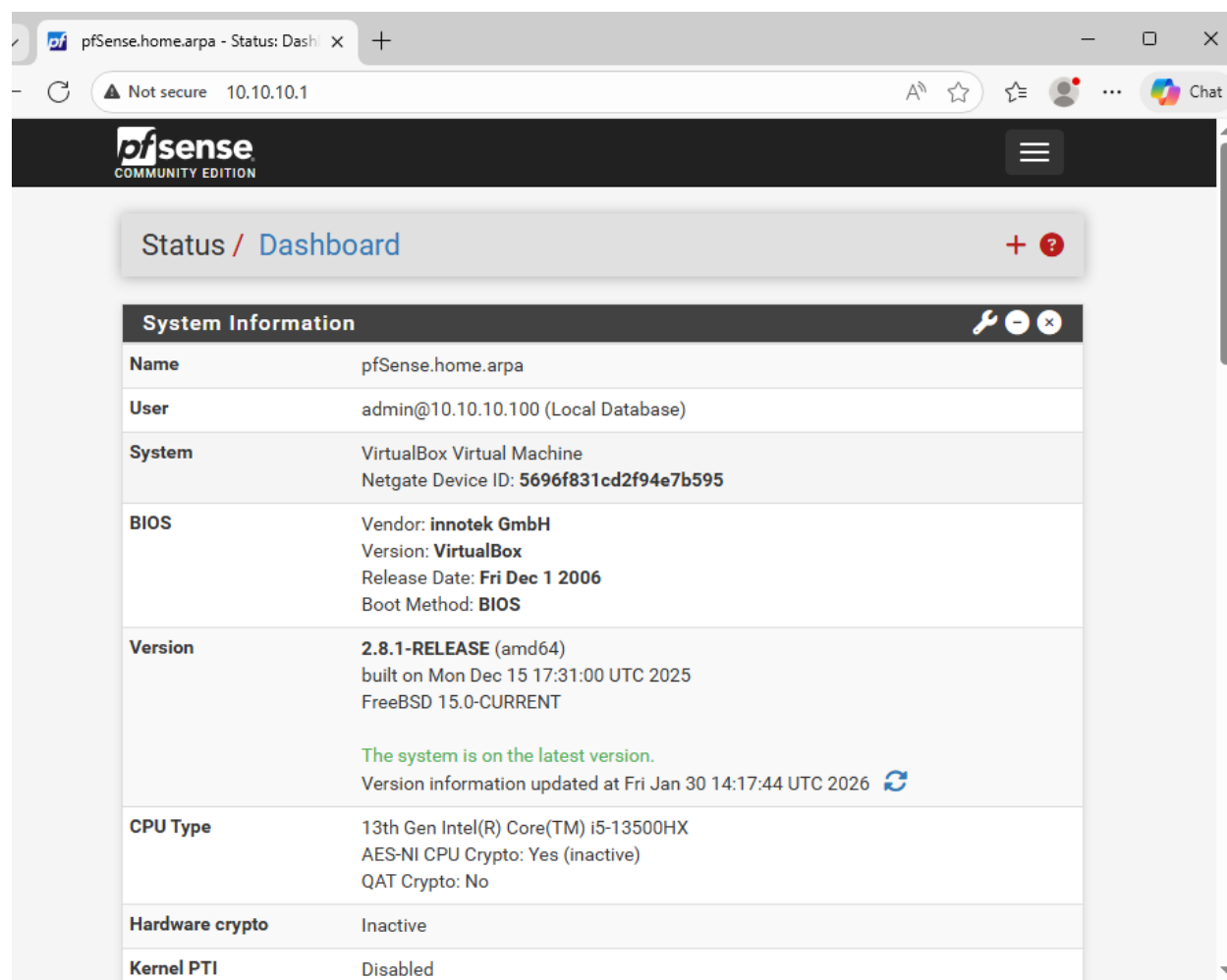
After installation, the pfSense web interface was accessed from Windows 10 browser:

URL: http://10.10.10.1

Default Credentials:

Username: admin

Password: pfsense



The screenshot shows a web browser window with the address bar displaying 'pfSense.home.arpa - Status: Dash' and '10.10.10.1'. The browser's address bar also shows 'Not secure'. The pfSense logo and 'COMMUNITY EDITION' are visible in the top left corner of the dashboard. The main content area shows the 'Status / Dashboard' breadcrumb and a 'System Information' panel. The 'System Information' panel contains the following details:

System Information	
Name	pfSense.home.arpa
User	admin@10.10.10.100 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 5696f831cd2f94e7b595
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Mon Dec 15 17:31:00 UTC 2025 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Fri Jan 30 14:17:44 UTC 2026
CPU Type	13th Gen Intel(R) Core(TM) i5-13500HX AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled

Task A: Block Specific Countries (GeoIP Blocking)

Step 6: Installing pfBlockerNG

pfBlockerNG package was installed to enable GeoIP and DNS-based blocking.

Installation Process:

7. Navigate to System → Package Manager → Available Packages
8. Search for "pfBlockerNG-devel"
9. Click Install and confirm
10. Wait for installation to complete
11. Access via Firewall → pfBlockerNG

pfBlockerNG-devel	3.2.10	pfBlockerNG-devel is the Next Generation of pfBlockerNG. Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats. GeoIP database by MaxMind Inc. (GeoLite2 Free version). De-Duplication, Suppression, and Reputation enhancements. Provision to download from diverse List formats. Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources. Domain Name (DNSBL) blocking via Unbound DNS Resolver. Package Dependencies: lighttpd-1.4.76 jq-1.7.1 gnugrep-3.11 rsync-3.4.0 py-maxminddb-2.6.2 libmaxminddb-1.12.2 iprange-1.0.4_2 grepcidr-2.0_1 python311-3.11.11 php83-8.3.19 php83-intl-8.3.19 py-sqlite3-3.11.11_8	+ Install
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Step 7: Configuring GeoIP Blocking

Configuration steps to block China and Russia:

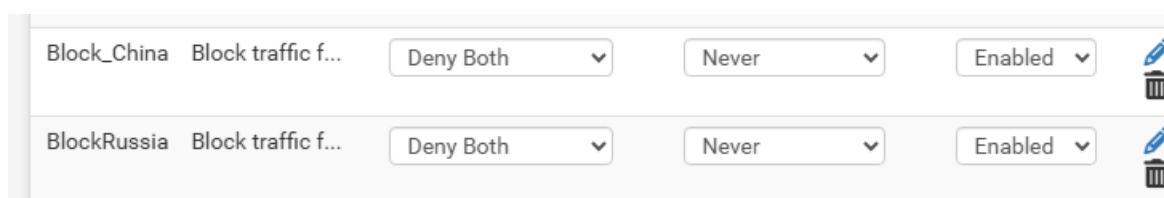
12. Navigate to Firewall → pfBlockerNG → IP
13. Click on IPv4 tab
14. Click "+ Add" to create new IPv4 rule
15. Set rule name: "Block_CN_RU"
16. Set Description: "Block China and Russia"
17. Set Action: "Deny Both" (blocks inbound and outbound)

18. Enable logging for monitoring purposes

Step 8: Adding Country IP Zones

IP zone files were obtained from [ipdeny.com](https://www.ipdeny.com):

Country	Zone File URL
China	https://www.ipdeny.com/ipblocks/data/countries/cn.zone
Russia	https://www.ipdeny.com/ipblocks/data/countries/ru.zone



Step 9: Testing GeoIP Blocking

To verify GeoIP blocking is working:

Test Method:

19. Open Command Prompt on Windows 10 client
20. Ping a Chinese IP address: ping 1.2.4.0
21. Ping a Russian IP address: ping 5.3.1.0
22. Both should timeout or show "Request timed out"
23. Check pfSense logs: Status → System Logs → Firewall
24. Verify blocked packets from China/Russia IPs

```

C:\Windows\system32>ping 223.5.5.5

Pinging 223.5.5.5 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 223.5.5.5:
    Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
Control-C
^C
C:\Windows\system32>ping 77.88.8.8

Pinging 77.88.8.8 with 32 bytes of data:
Request timed out.
Request timed out.

Ping statistics for 77.88.8.8:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),
Control-C
^C
C:\Windows\system32>ping 1.1.1.1

Pinging 1.1.1.1 with 32 bytes of data:
Reply from 1.1.1.1: bytes=32 time=49ms TTL=254
Reply from 1.1.1.1: bytes=32 time=50ms TTL=254
Reply from 1.1.1.1: bytes=32 time=50ms TTL=254

Ping statistics for 1.1.1.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),

```

Blocked Ips:

General
IP
DNSBL
Update
Reports
Feeds
Logs
Sync

Log/File Browser selections

Log/File type: Log Files
Choose which type of log/file you want to view.

Log/File selection: ip_block.log
Choose which log/file you want to view.

Log/File Details

File successfully loaded: Total Lines: 51
Log/File Path: /var/log/pfblockerlog/ip_block.log

Log

```

Jan 28 11:56:10,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 11:56:10,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 11:56:25,1770008580,em1,LAN,block,4,1,ICMP,10.10.10.100,5.8.0.0,request,1,out,Unk,pfB_BlockRuss:
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 13:36:34,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 14:10:36,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 14:10:36,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 14:10:36,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir
Jan 28 14:10:36,1770008527,em1,LAN,block,4,1,ICMP,10.10.10.100,1.2.4.0,request,1,out,Unk,pfB_Block_Chir

```

Task B: Restrict Access to Specified Websites

Step 10: Configuring DNSBL (DNS Blacklist)

DNS-based blocking was configured to restrict access to social media websites.

Configuration Steps:

25. Navigate to Firewall → pfBlockerNG → DNSBL
26. Enable DNSBL functionality
27. Click on "DNSBL Feeds" tab
28. Create custom blacklist: Click "+ Add"
29. Set name: "Social_Media_Block"
30. Set Action: "Deny"

Step 11: Adding Blocked Domains

The following domains were added to the blacklist:

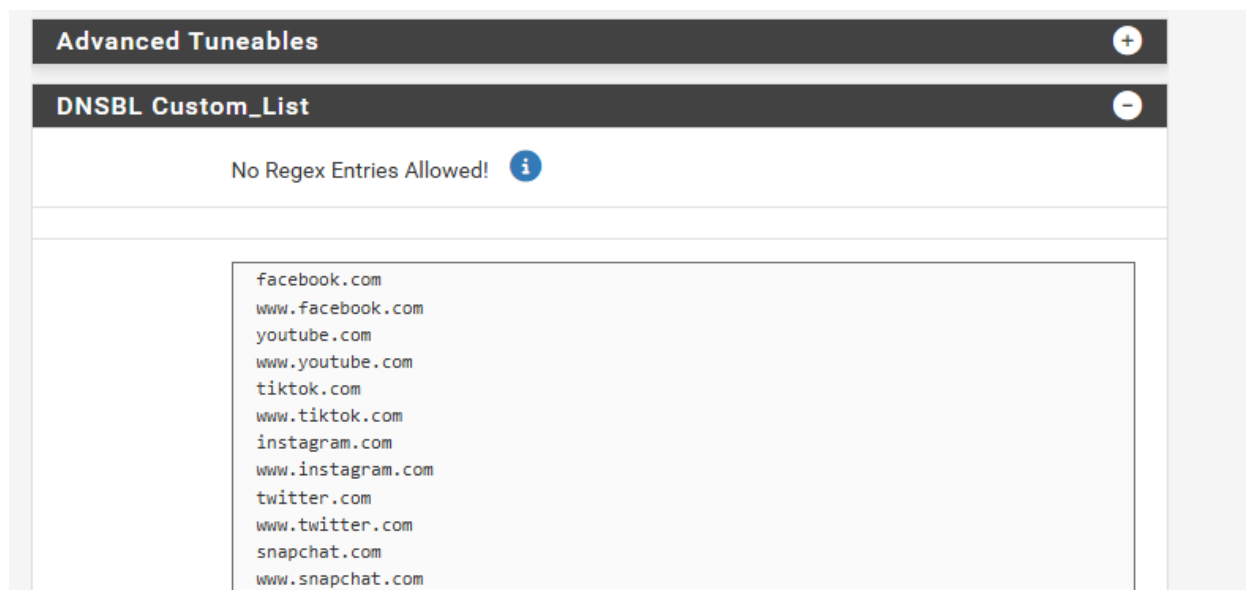
- facebook.com and *.facebook.com
- instagram.com and *.instagram.com
- twitter.com and *.twitter.com
- x.com and *.x.com

Note: The asterisk (*) wildcard ensures all subdomains are also blocked (e.g., www.facebook.com, m.facebook.com, etc.)

Info

Links
Firewall Aliases
Firewall Rules
Firewall Logs

Name / Description	Blocked_Websites	Block social media and non-work sites
Enter Name and Description. i		



Step 12: Configuring DNS Resolver

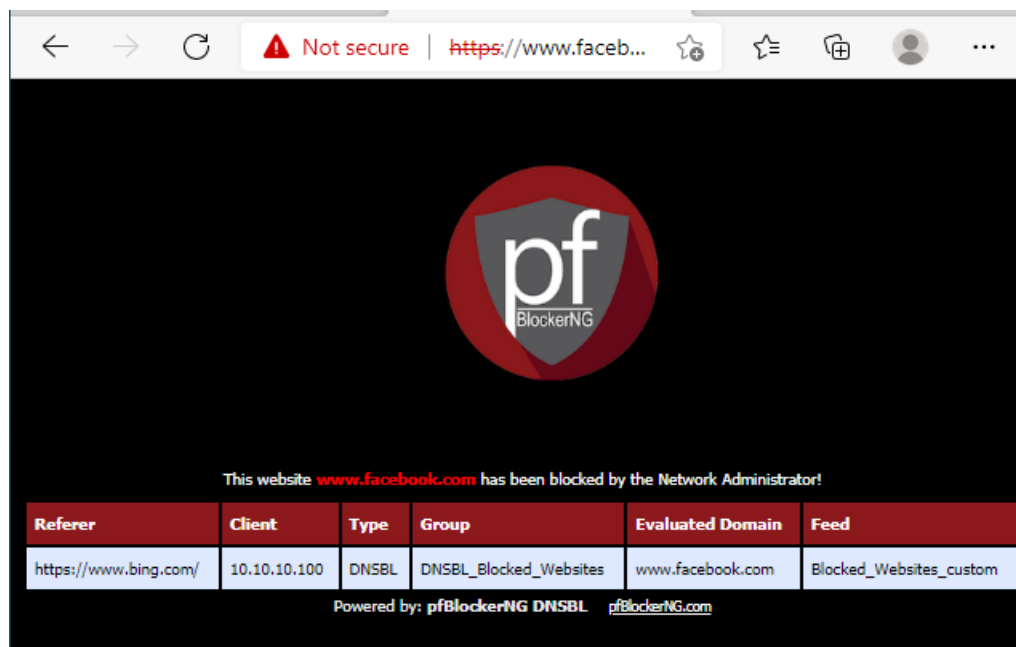
DNS resolver must be configured to use pfBlockerNG:

31. Navigate to Services → DNS Resolver
32. Ensure DNS Resolver is enabled
33. Under "General Settings", verify pfBlockerNG integration
34. Set DNSBL mode to "python" (better subdomain handling)
35. Apply changes and restart DNS Resolver service

Step 13: Testing Website Blocking

To verify website blocking:

36. Open web browser on Windows 10 client
37. Try accessing facebook.com
38. Try accessing instagram.com
39. Try accessing twitter.com/x.com
40. All should be blocked with pfBlockerNG block page



pfBlockerNG DNSBL logs:

```
DNSBL-Full,Jan 28 11:32:31,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:32:36,c.bing.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,c.bing.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:32:41,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:03,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:13,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:18,facebook.com,10.10.10.100,-|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Gecko) Chrome/92.0.4515.
DNSBL-1x1,Jan 28 11:44:25,facebook.com,10.10.10.100,-|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Gecko) Chrome/92.0.4515.
DNSBL-Full,Jan 28 11:44:25,facebook.com,10.10.10.100,-|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Gecko) Chrome/92.0.4515.
DNSBL-Full,Jan 28 11:44:25,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:25,c.bing.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,c.bing.com,StevenBlack_Ads,+
DNSBL-1x1,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET /favicon.ico HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML
DNSBL-Full,Jan 28 11:44:32,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Geck
DNSBL-1x1,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Geck
DNSBL-Full,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Geck
DNSBL-1x1,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Geck
DNSBL-Full,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Geck
DNSBL-1x1,Jan 28 11:44:32,facebook.com,10.10.10.100,http://facebook.com/|GET / HTTP/1.1|Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Geck
DNSBL-Full,Jan 28 11:44:32,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:32,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:37,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,-
DNSBL-Full,Jan 28 11:44:37,c.bing.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,c.bing.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:37,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:46,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,-
DNSBL-Full,Jan 28 11:44:46,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:50,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,-
DNSBL-Full,Jan 28 11:44:50,c.bing.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,c.bing.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:44:50,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,+
DNSBL-Full,Jan 28 11:45:25,srtb.msn.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,srtb.msn.com,StevenBlack_Ads,-
DNSBL-Full,Jan 28 11:45:25,sb.scorecardresearch.com,10.10.10.100,-|PRI|HTTP/2.0|-|DNSBL,DNSBL_Ads_Basic,sb.scorecardresearch.com,StevenBlack_Ads,+
```

Task C: Create Administrator Privilege Rules

Step 14: Creating Admin IP Alias

An IP alias was created to define authorized administrator IP addresses.

Configuration Steps:

41. Navigate to Firewall → Aliases
42. Click on "IP" tab
43. Click "+ Add" to create new alias
44. Set Name: "AdminIPs"
45. Set Description: "Authorized administrator IP addresses"
46. Set Type: "Host(s)"
47. Add IP addresses of authorized administrators
48. Click Save and Apply Changes

Firewall / Aliases / Edit ?

Properties

Name
The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".

Description
A description may be entered here for administrative reference (not parsed).

Type

Host(s)

Hint Enter as many hosts as desired. Hosts must be specified by their IP address or fully qualified domain name (FQDN). FQDN hostnames are periodically re-resolved and updated. If multiple IPs are returned by a DNS query, all are used. An IP range such as 192.168.1.1-192.168.1.10 or a small subnet such as 192.168.1.16/28 may also be entered and a list of individual IP addresses will be generated.

IP or FQDN

Step 15: Configuring Admin Access Rules

Firewall rules were created on the LAN interface:

Rule #	Source	Destination	Action
1	AdminIPs	pfSense:443	Allow
2	Any	pfSense:443	Block
3	LAN net	Any	Allow

Rule Processing: Firewall rules are processed top-to-bottom. The first matching rule is applied. This configuration ensures only AdminIPs can access the web interface.

Firewall / Rules / LAN ⋮ ⌂ ?											
Floating WAN <u>LAN</u>											
Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	5/1.44 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	0/0 B	IPv4 *	*	*	pfB_PRI1_v4	*	*	none		pfB_PRI1_v4	
☐	0/0 B	IPv4 *	*	*	pfB_Block_China_v4	*	*	none		pfB_Block_China_v4	
☐	0/0 B	IPv4 *	*	*	pfB_BlockRussia_v4	*	*	none		pfB_BlockRussia_v4	
☐	0/260 B	IPv4 TCP	*	*	This Firewall (self)	*	*	none		Block admin access from unauthorized IPs	
☐	0/0 B	IPv4 TCP	AdminPCs	*	This Firewall (self)	443 (HTTPS)	*	none		Allow admin access from authorized IPs	
☐	38/209.66 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Step 16: Testing Admin Access Control

To verify the administrator access rules:

Test from Authorized IP:

- Access pfSense dashboard from AdminIPs address
- Should successfully load <https://10.10.10.1>
- Login should work normally

Test from Unauthorized IP:

- Access pfSense dashboard from non-AdminIPs address
- Connection should timeout or be refused
- Check firewall logs to confirm blocking

← → ↻ ⚠ Not secure | 10.10.10.1

pfSense COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Status / Dashboard + ?

System Information

Name	pfSense.home.arpa
User	admin@10.10.10.100 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: 5696f831cd2f94e7b595
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Mon Dec 15 17:31:00 UTC 2025 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Sun Feb 1 14:36:25 UTC 2026
CPU Type	13th Gen Intel(R) Core(TM) i5-13500HX AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 24 Minutes 12 Seconds
Current date/time	Sun Feb 1 14:59:08 UTC 2026
DNS server(s)	• 127.0.0.1 • ::1 • 192.168.1.1
Last config change	Sat Jan 31 8:04:12 UTC 2026
State table size	0% (111/199000) Show states
MBUF Usage	6% (1055/1638400)

Netgate Services And Support

Contract type **Community Support**
Community Support Only

NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the [NETGATE RESOURCE LIBRARY](#).

You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.

- [Upgrade Your Support](#)
- [Community Support Resources](#)
- [Netgate Global Support FAQ](#)
- [Official pfSense Training by Netgate](#)
- [Netgate Professional Services](#)
- [Visit Netgate.com](#)

If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your **Netgate Device ID (NDI)** from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase TAC supports [here](#).

Interfaces

WAN	↑ 1000baseT <full-duplex>	10.0.2.15 fd17:625c:f037:2:a00:27ff:fed:a054
LAN	↑ 1000baseT <full-duplex>	10.10.10.1

pfBlockerNG

Blocked access from unauthorized IP

10.10.10.1

10.10.10.1

Hmmm... can't reach this page

10.10.10.1 took too long to respond

Try:

- [Checking the connection](#)
- [Checking the proxy and the firewall](#)
- [Running Windows Network Diagnostics](#)

ERR_CONNECTION_TIMED_OUT

Details

Administrator: Command Prompt

```

Microsoft Windows [Version 10.0.19045.2865]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::9490:1cca:325f:756e%10
    IPv4 Address. . . . . : 10.10.10.150
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.10.10.1

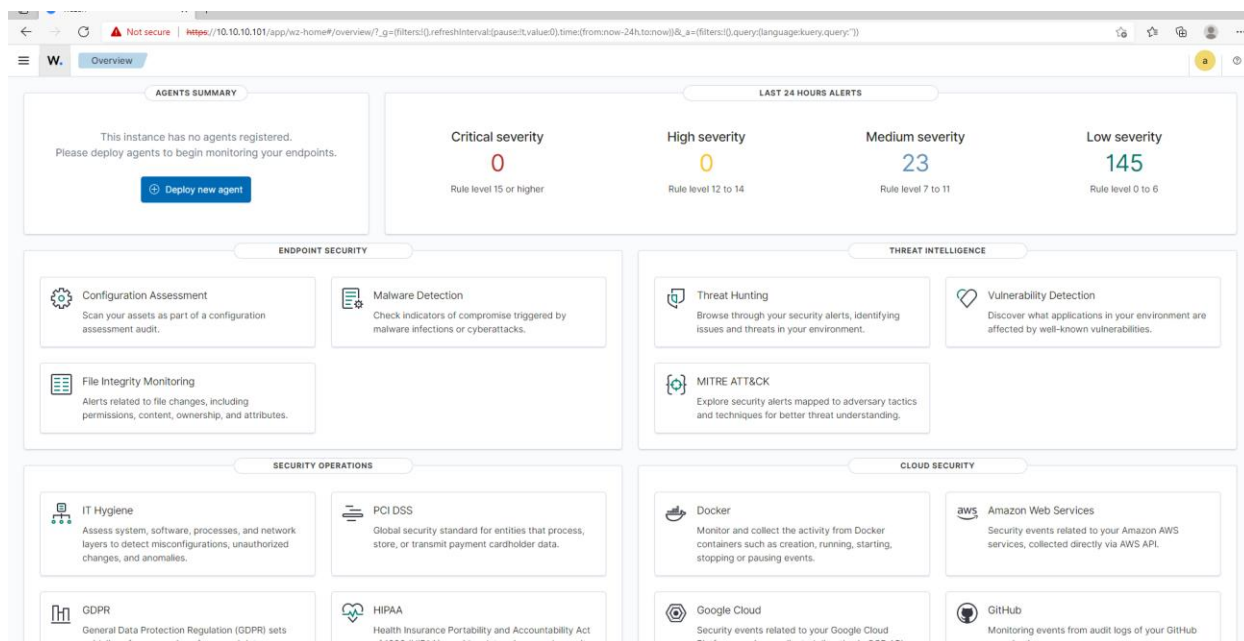
C:\WINDOWS\system32>

```

The Wazuh web interface can be accessed from any browser on the LAN:

URL: <https://10.10.10.101>

Login: Use credentials from installation



Step 19: Deploying Wazuh Agent on Windows

Wazuh agent was installed on Windows 10 client for monitoring.

Agent Deployment:

55. Navigate to Wazuh Dashboard → Agents
56. Click "Deploy new agent"
57. Select Windows as operating system
58. Set Wazuh server address: 10.10.10.101
59. Set agent name: WIN10-CLIENT
60. Copy the installation command
61. Run command in PowerShell as Administrator
62. Start the Wazuh agent service

W. Endpoints Deploy new agent a ?

< Deploy new agent

✓ Select the package to download and install on your system:

LINUX

☐ RPM amd64
 ☐ RPM aarch64
 ☐ DEB amd64
 ☐ DEB aarch64

WINDOWS

☒ MSI 32/64 bits

macOS

☐ Intel
 ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

✓ Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ?

10.10.10.101

☐ Remember server address

✓ Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: ?

WIN10-CLIENT

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\WINDOWS\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.1-1.msi -OutFile $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='10.10.10.101' WAZUH_AGENT_NAME='WIN10-CLIENT'

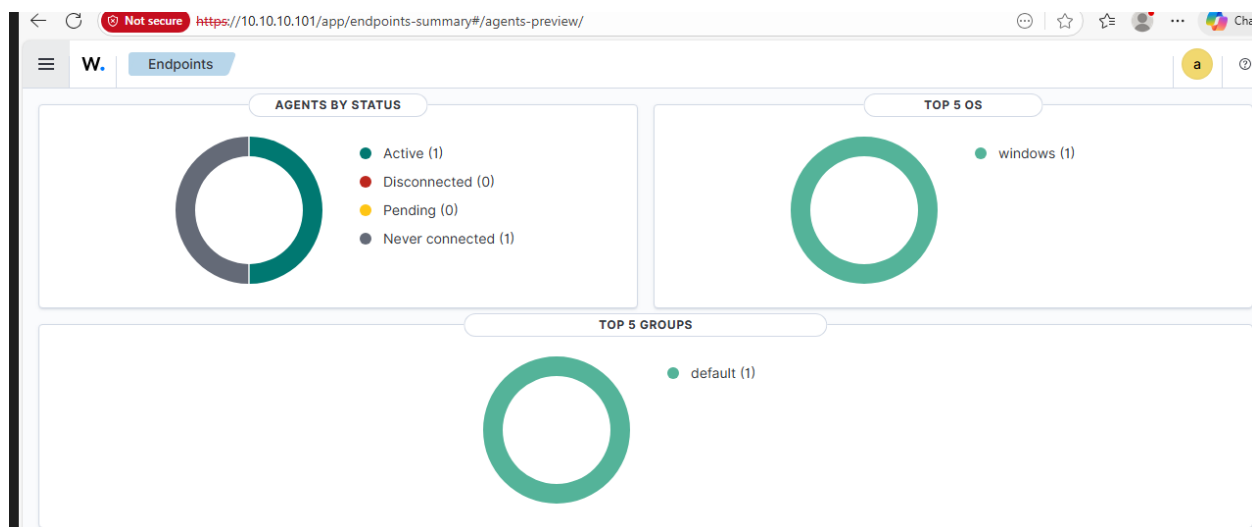
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> NET START Wazuh

The Wazuh service was started successfully.
```

Step 20: Verifying Agent Connection

To confirm the agent is properly connected:

63. Return to Wazuh Dashboard
64. Navigate to Agents Management
65. Look for WIN10-CLIENT in agent list
66. Status should show "Active" with green indicator
67. Check "Last Keep Alive" timestamp (should be recent)



Step 21: Integrating pfSense Logs with Wazuh

pfSense was configured to forward logs to Wazuh server via syslog.

pfSense Configuration:

68. Navigate to Status → System Logs → Settings
69. Scroll to "Remote Logging Options"
70. Check "Enable Remote Logging"
71. Set Remote log servers: 10.10.10.101:514
72. Select log contents: System Events, Firewall Events, DHCP Events
73. Set Remote Syslog Contents: Everything
74. Click Save

Remote Logging Options

Enable Remote Logging

☒ Send log messages to remote syslog server

Source Address

LAN

This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If a single IP is picked, remote syslog servers must all be of that IP type. To mix IPv4 and IPv6 remote syslog servers, bind to all interfaces.

NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.

IP Protocol

IPv4

This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; If an IP address of the selected type is not found on the chosen interface, the other type will be tried.

Remote log servers

10.10.10.101:514

IP[:port]

IP[:port]

Remote Syslog Contents

☒ Everything

☐ System Events

☐ Firewall Events

☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns)

☐ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client)

☐ PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client)

☐ General Authentication Events

☐ Captive Portal Events

☐ VPN Events (IPsec, OpenVPN, L2TP, PPPoE Server)

☐ Gateway Monitor Events

☐ Routing Daemon Events (RADVD, UPnP, RIP, OSPF, BGP)

☐ Network Time Protocol Events (NTP Daemon, NTP Client)

☐ Wireless Events (hostapd)

Syslog sends UDP datagrams to port 514 on the specified remote syslog server, unless another port is specified. Be sure to set syslogd on the remote server to accept syslog messages from pfSense.

Step 22: Configuring Wazuh to Receive Syslog

Wazuh manager was configured to listen for syslog messages.

The remote syslog configuration was added to ossec.conf and the Wazuh manager was restarted.

```
[wazuh-user@wazuh-server ~]$ sudo netstat -anu | grep 514
udp        0      0 0.0.0.0:514          0.0.0.0:*
[wazuh-user@wazuh-server ~]$
```

Step 23: Verifying Log Reception

To confirm pfSense logs are being received by Wazuh:

75. Navigate to Wazuh Dashboard → Explore → Discover
76. Set time range to "Last 15 minutes"
77. Search for: agent.ip: 10.10.10.1
78. Should see pfSense firewall logs appearing

79. Verify log fields contain firewall data

80. Check for blocked connection logs from GeoIP rules

	Feb 2 10:53:00	LAN	 pfB_Block_China_v4 (1770008527)	 10.10.10.100	 223.5.5.5	ICMP
	Feb 2 10:53:05	LAN	 pfB_Block_China_v4 (1770008527)	 10.10.10.100	 223.5.5.5	ICMP
	Feb 2 10:53:05	LAN	Default deny rule IPv4 (1000000103)	 10.10.10.100:49881	 2.16.158.185:443	TCP:FA
	Feb 2 10:53:10	LAN	Default deny rule IPv4 (1000000103)	 10.10.10.100:49882	 2.16.158.185:443	TCP:FA
	Feb 2 10:53:10	LAN	 pfB_Block_China_v4 (1770008527)	 10.10.10.100	 223.5.5.5	ICMP
	Feb 2 10:53:15	LAN	 pfB_Block_China_v4 (1770008527)	 10.10.10.100	 223.5.5.5	ICMP
	Feb 2 10:53:20	LAN	Default deny rule IPv4 (1000000103)	 10.10.10.100:49886	 2.16.158.51:443	TCP:FA
	Feb 2 10:53:22	LAN	Default deny rule IPv4 (1000000103)	 10.10.10.100:49889	 2.16.158.185:443	TCP:FA
	Feb 2 10:53:23	LAN	Default deny rule IPv4 (1000000103)	 10.10.10.100:49887	 2.16.158.51:443	TCP:FA

File Integrity Monitoring (FIM)

Step 24: Configuring FIM on Windows Agent

File Integrity Monitoring was configured to detect unauthorized file changes on the Windows client.

The ossec.conf file was edited to include syscheck configuration for real-time monitoring of critical directories.

Monitored Directories:

- C:\Program Files (real-time)
- C:\Program Files (x86) (real-time)
- C:\Windows\System32 (real-time)
- C:\Users\Public (real-time)
- C:\Users (periodic scanning)

```
//fim//
<syscheck>
<!-- Frequency of scans (in seconds) -->
<scan_on_start>yes</scan_on_start>

<!-- Real-time monitoring (immediate alerts) -->
<directories realtime="yes">C:\Program Files</directories>
<directories realtime="yes">C:\Program Files (x86)</directories>
<directories realtime="yes">C:\Windows\System32</directories>
<directories realtime="yes">C:\Users\Public</directories>

<!-- Periodic scanning every 6 hours (less CPU) -->
<directories check_all="yes" report_changes="yes" scan_time="6h">C:\Users</directories>

<!-- Exclusions to reduce noise -->
<ignore>C:\Users\*\AppData\Local\Temp</ignore>
<ignore>C:\Windows\Temp</ignore>
<ignore>C:\Windows\Logs</ignore>
<ignore>C:\ProgramData\Microsoft\Windows\WER</ignore>
</syscheck>
```

FIM Configuration Optimization

The configuration was optimized to balance security and performance:

Strategy	Implementation
Real-time Monitoring	System32, Program Files - immediate detection
Periodic Scanning	User directories - reduced CPU load
Exclusions	Temporary files, logs - reduced noise

Step 25: Testing File Integrity Monitoring

FIM was tested by creating, modifying, and deleting files.

Test Results:

Action	Test Command	Wazuh Alert
File Created	echo > test.txt	Rule 554 ✓
File Modified	echo >> test.txt	Rule 550 ✓
File Deleted	del test.txt	Rule 553 ✓
Permissions Changed	icaccls modify	Rule 555 ✓

```

:\WINDOWS\system32>net stop wazuh

The Wazuh service was stopped successfully.

:\WINDOWS\system32>net start wazuh

The Wazuh service was started successfully.

:\WINDOWS\system32>echo TEST FIM > "C:\Program Files (x86)\test_fim.txt"

:\WINDOWS\system32>echo MODIFIED >> "C:\Program Files (x86)\test_fim.txt"

:\WINDOWS\system32>del "C:\Program Files (x86)\test_fim.txt"

:\WINDOWS\system32>mkdir "C:\Windows\System32\test_fim_dir" & rmdir "C:\Windows\System32\test_fim_dir"

```

	Feb 2, 2026 @ 11:45:47.893	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:47.574	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-logcollector....	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:42.851	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:37.834	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:32.830	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:27.826	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:23.023	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:20.097	DESKTOP-JRIPB6J	c:\users\public\fim-test.txt	added	File added ...	5
	Feb 2, 2026 @ 11:45:17.939	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:45:13.751	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\ossec.log	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:42:45.956	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:42:40.883	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:42:36.307	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\wazuh-agent.state	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:42:34.255	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\ossec.conf	modified	Integrity ch...	7
	Feb 2, 2026 @ 11:42:33.623	DESKTOP-JRIPB6J	c:\program files (x86)\ossec-agent\ossec.conf	modified	Integrity ch...	7

Step 26: Understanding FIM Alert Rules

Wazuh generates different alerts based on the type of file change:

Rule ID	Severity Level	Description
550	7	Integrity checksum changed
553	7	File deleted from system
554	5	File added to system
555	7	File permissions modified
556	7	File owner modified

Security Monitoring Results

Step 27: Accessing Security Events in Wazuh

Security events can be viewed in the Wazuh Dashboard.

Navigation: Explore → Discover

Useful Search Queries:

- syscheck - View all File Integrity Monitoring alerts
- 10.10.10.1 - View all pfSense firewall logs
- rule.groups:syscheck - Filter specifically for FIM events
- rule.level:>=7 - View high severity alerts
- rule.groups:firewall - View firewall-related events

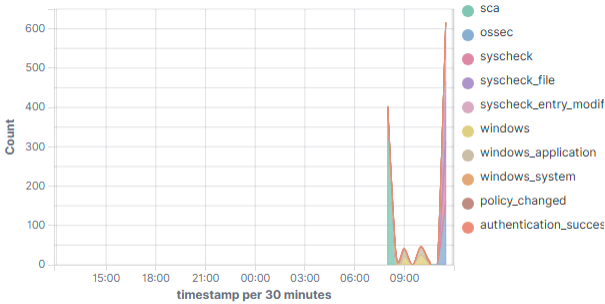
Step 28: Summary of Blocked Traffic

Category	Description	Status
GeoIP Blocks	China, Russia traffic blocked	✓ Working
Website Blocks	Social media sites blocked	✓ Working
Admin Access	Unauthorized access denied	✓ Working
FIM Alerts	File changes detected	✓ Working

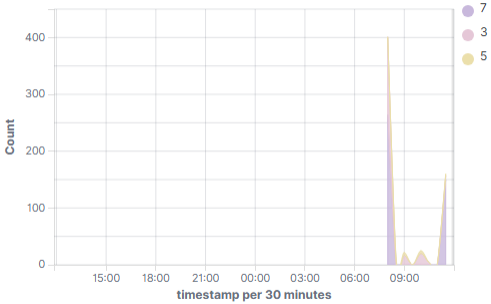
Step 29: Security Events Observed

Event Type	Count	Severity
GeoIP Blocked Traffic	Multiple	High
Website Access Denied	Multiple	Medium
File Integrity Changes	Multiple	Medium
Admin Login Attempts	Few	Low

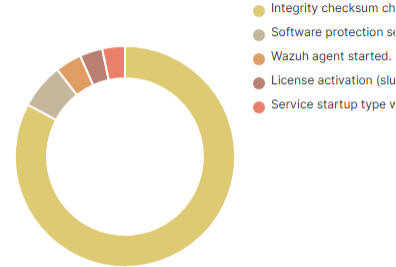
Top 10 Alert groups evolution



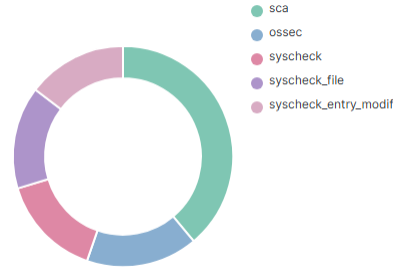
Alerts



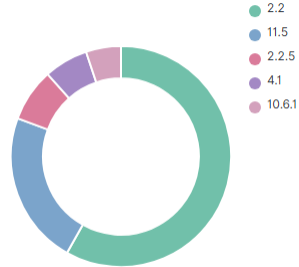
Top 5 alerts



Top 5 rule groups



Top 5 PCI DSS Requirements



Results and Observations

The following results were achieved through this implementation:

- ✓ pfSense successfully deployed as network gateway
- ✓ GeoIP blocking effectively prevents connections to/from blocked countries
- ✓ DNSBL successfully blocks access to specified websites
- ✓ Admin rules restrict dashboard access to authorized IPs
- ✓ pfSense logs successfully sent to Wazuh via syslog
- ✓ Wazuh agent on Windows reporting file integrity changes
- ✓ Real-time FIM alerts generated for file modifications
- ✓ False positives reduced through proper tuning

Challenges Faced and Solutions

Challenge	Solution
Network conflict (WAN/LAN same subnet)	Changed LAN to 10.10.10.0/24
pfSense not sending logs	Restarted syslogd service
Wazuh agent disconnecting	Set static IP on Windows
FIM not generating alerts	Simplified syscheck config
Real-time monitoring not working	Reduced monitored directories

Recommendations for Future Enhancement

1. Expand Geo-Blocking

- Add more high-risk countries (North Korea, Iran, Syria)
- Regularly update GeoIP databases
- Consider blocking entire regions for highly sensitive environments

2. Scheduled Updates

- Configure automatic pfBlockerNG updates (weekly)
- Keep Wazuh agents updated
- Automate GeoIP database refresh

3. Alert Configuration

- Set up email alerts for high-severity events
- Configure thresholds for repeated attacks
- Integrate with ticketing system for incident tracking
- Create custom alert rules for specific threats

4. Additional Security Layers

- Implement IDS/IPS (Snort or Suricata)
- Enable SSL/TLS inspection
- Add two-factor authentication for admin access
- Deploy honeypots for threat intelligence

5. Log Retention and Management

- Increase log retention to 90 days minimum
- Configure log rotation policies
- Implement log archiving to external storage
- Set up log backup for compliance requirements

6. FIM Enhancements

- Monitor additional critical directories
- Add more registry keys to monitoring
- Fine-tune ignore patterns to reduce noise
- Implement baseline comparison for anomaly detection

Conclusion

This task successfully implemented a comprehensive security monitoring solution that combines pfSense firewall with Wazuh SIEM for centralized visibility and control.

Key Capabilities Achieved:

- ✓ Software-based firewall (pfSense) deployed and configured
- ✓ Geographic IP blocking prevents traffic from high-risk countries
- ✓ Website access control using DNS-based blacklisting
- ✓ Administrative access restricted to authorized IPs only
- ✓ Centralized log monitoring with Wazuh SIEM
- ✓ File Integrity Monitoring detects unauthorized file changes
- ✓ Real-time security event visibility achieved

The integration of pfSense firewall with Wazuh SIEM provides a robust security infrastructure capable of detecting and blocking threats while maintaining visibility through centralized logging and alerting. The File Integrity Monitoring adds an additional layer of security by detecting unauthorized changes to critical system files and registry keys.

This setup establishes a strong foundation for advanced security monitoring, threat detection, and incident response in enterprise environments. The modular nature of the solution allows for easy expansion and integration with additional security tools as organizational needs evolve.

Appendix

A. Tools and Versions Used

Tool	Version
VirtualBox	
pfSense	Community Edition
pfBlockerNG	devel
Wazuh Server	4.x
Wazuh Agent	4.7.0
Windows 10	

B. Network Configuration Details

- LAN Subnet: 10.10.10.0/24
- Gateway: 10.10.10.1
- DHCP Range: 10.10.10.100 - 10.10.10.200
- DNS: 10.10.10.1
- Syslog Port: 514 (UDP)

C. IP Zone Files Used

China: <https://www.ipdeny.com/ipblocks/data/countries/cn.zone>

Russia: <https://www.ipdeny.com/ipblocks/data/countries/ru.zone>

D. References and Documentation

- pfSense Documentation: <https://docs.netgate.com/pfsense/>
- Wazuh Documentation: <https://documentation.wazuh.com/>
- pfBlockerNG Wiki: <https://docs.netgate.com/pfsense/en/latest/packages/pfblocker.html>
- IPDeny GeoIP Database: <https://www.ipdeny.com/ipblocks/>